

Fly Methods

Results

Markov Chains

A Markov chain is a probabilistic model of discrete states. Given a set of n possible states, the Markov chain is typically defined as an $n \times n$ adjacency matrix where p_{ij} is the transition from state i to state j .

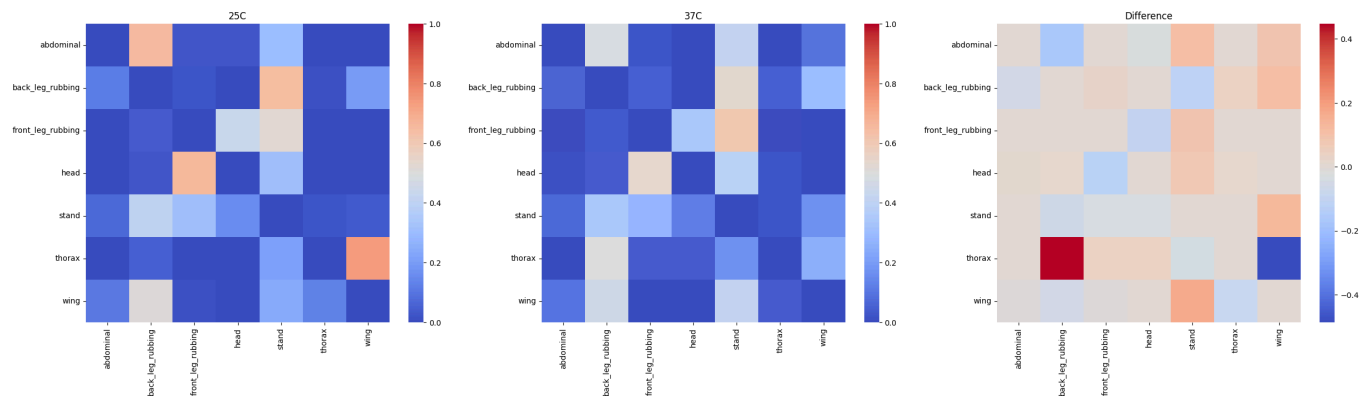
$$\begin{bmatrix} p_{11} & \dots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \dots & p_{nn} \end{bmatrix}$$

They are a natural way to describe discrete fly behavior and have been well-documented in the literature ([Mueller et. al. 2019](#)). Technically, there are two parts to a Markov chain modeling approach: the matrix and starting probabilities. In this case, we don't really care about the initial states since there is no real $t = 0$, but this could be applicable in future cases.

We have a maximum likelihood estimate for the transition probability between two states.

$$\hat{P}_{ij} = \frac{n_{ij}}{\sum_{k=1}^n n_{ik}}$$

Using that estimate, we can do a lot of analysis. The *MarkovChain* object is a class that I created to do most of this analysis, taking a CSV of behavioral data and converting it into matrices for each fly. Even just visualizing and comparing matrices is useful, for example the following diagram of heat-stressed flies.

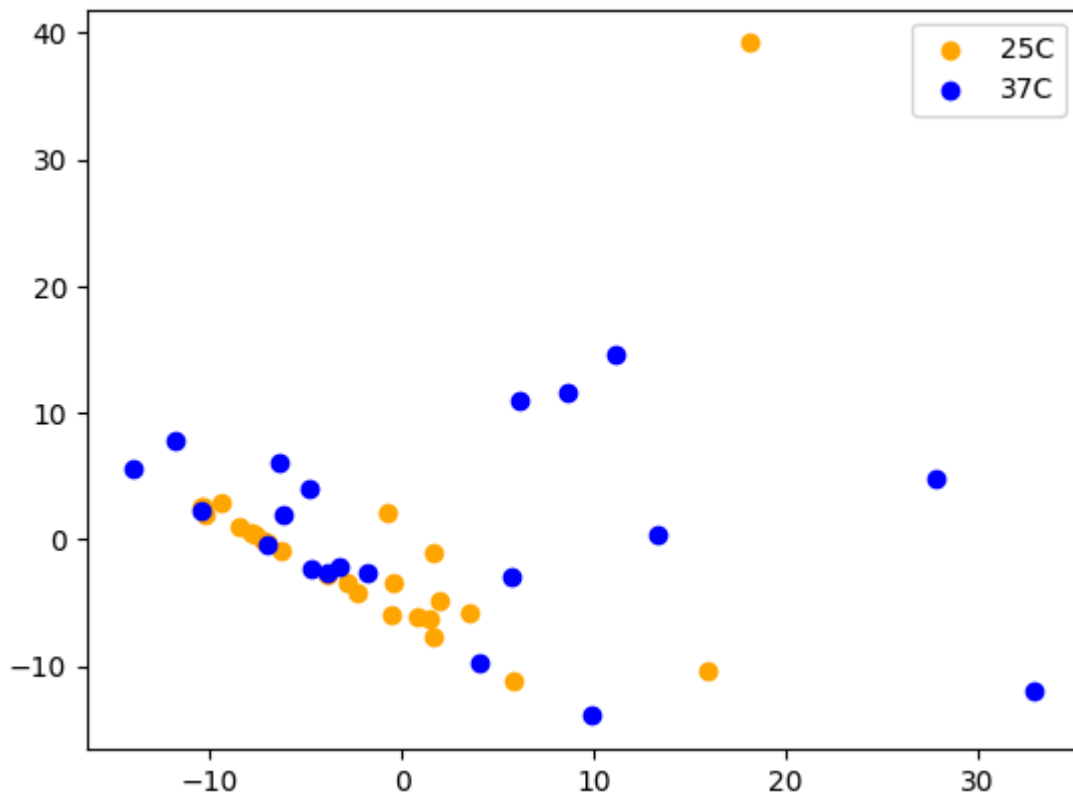


It's unclear to me how best to summarize differences between populations. In this case I calculated each of the matrices by computing a population-level transition matrix since the row probabilities won't add up to 1 if we just average over the matrices.

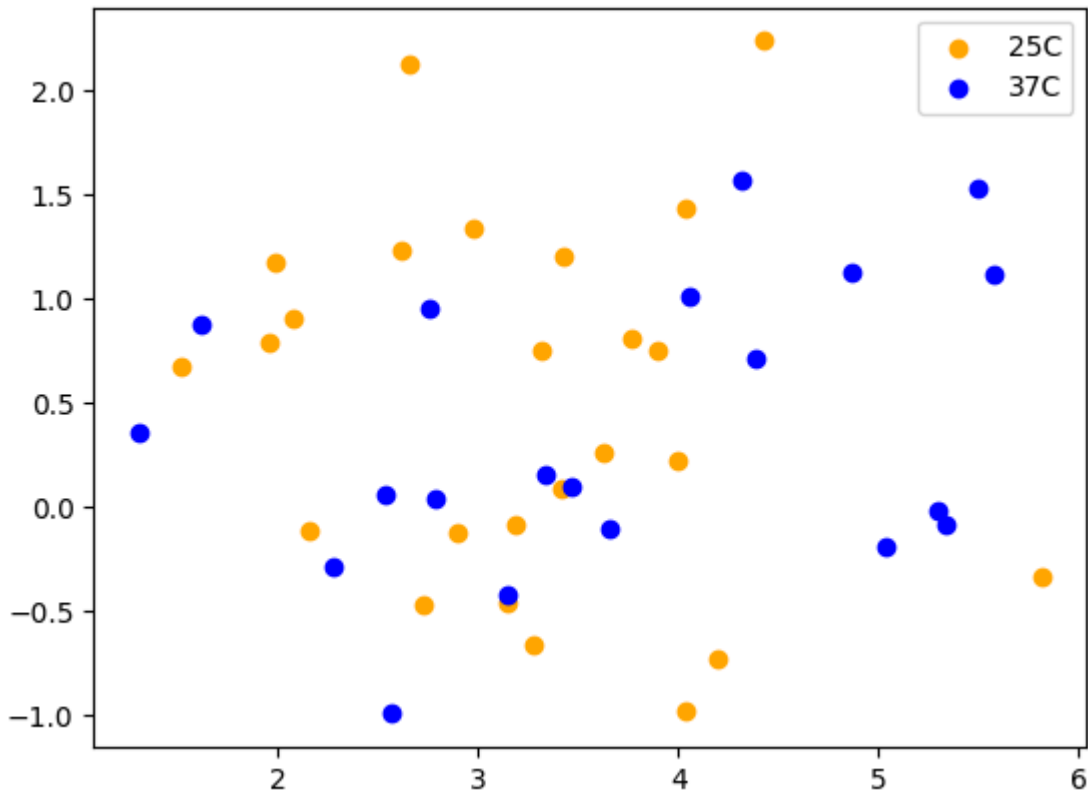
Clustering

[Mueller et. al. 2021](#) showed a number of potentially useful methods for doing Markov chain-based analysis. They excluded self-transitions, resulting in a 42 dimensional vector based on their 7 states ($7 * 7 - 7 = 42$). This high-dimensional vector space isn't very interpretable, so they used dimensionality reduction techniques (t-SNE) to plot their points in 2D space.

There are two main techniques that are used to do this task that are commonly accepted in research right now (2026): t-Stochastic Neighbor Embedding (t-SNE) and Principal Component Analysis (PCA). They differ in important ways.



Alternatively, we can use t-SNE, which iteratively minimizes the entropy between the high and low-dimensional distributions.



Some researchers believe that more ML-based approaches like t-SNE and UMAP are biased. Along with the classic problem of [the Garden of Forking Paths](#) associated with choosing analyses, these types dimensionality reduction techniques don't necessarily preserve relationships between points and can introduce [spurious patterns in some cases](#). In this case, I would say that neither method really showed anything, but they're definitely valuable techniques to explore.

Hypothesis Testing

In statistical hypothesis testing, we seek to determine whether a result is consistent with the null hypothesis, or in a treatment-based setting, the control distribution.

Chi-square tests measure a difference in proportions between two sets of data. In this case, the proportions are the number of transitions for each transition over the total number of transitions. There are a few necessary assumptions, namely that there are observed values of each transition, so this is not the most highly recommended method. The formula is given by [Billingsley](#), taking a null transition probability matrix P_0 and estimated matrix $\hat{P}$ as well as its frequencies N and the number of transitions k .

$$\sum_i \sum_j \frac{n_{ij}(\hat{p}_{ij} - p_{ij}^0)^2}{n_i p_{ij}^0} \sim \chi^2(k(k-1))$$

Again this isn't very interpretable and it's prone to differences in individual probabilities.

Entropy

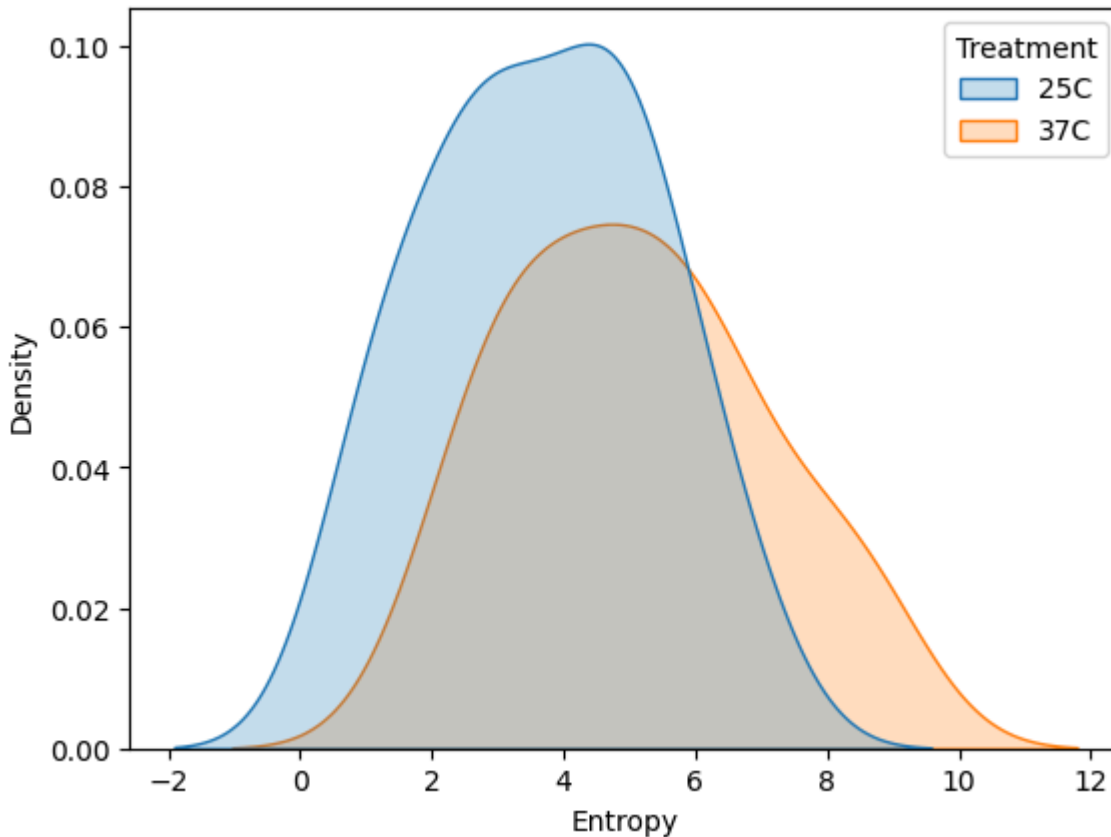
Entropy is the degree of "surprise" in a probability distribution. In information theory (not physics, it's given by the following formula for a random variable X .

$$H(X) := - \sum_{x \in X} p(x) \log p(x)$$

Although I didn't know it when I first tried it out (I referred to [Vegetabile et. al. 2019](#)), Mueller et. al 2021 calculates the "entropy rate" for a Markov chain. This is given by

$$H = \sum_{ij} \mu_i P_{ij} \log P_{ij}$$

where μ is the steady-state distribution. Having gathered n values of H for the n members of each population, we can perform normal statistical inference, using either parametric or non-parametric methods. In the following, we show the heat stress entropies:



This certainly looks like a difference in normal distributions, so we can easily run a t-test and obtain

$$t^* = -2.672$$
$$p = 0.005$$

for the hypothesis that the average entropy of the 37C distribution is higher, meaning the actions of the flies are more "surprising".

| Future Methods

| Hidden Markov Models

Hidden Markov Models (HMMs) are powerful tools for processing sequential data. They work by computing the maximum likelihood sequence of latent states that produce the actual data. In order to perform inference, we need two inputs: the latent transition matrix and the emission function.

$$\begin{bmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \cdots & p_{nn} \end{bmatrix}$$

is the transition matrix, which is distinct from the observable Markov transition matrices we have above. We also need an emission function:

$$P(\text{observed emission} | \text{latent state})$$

I plan to use the *hmmlearn* Python library, which supports both approaches to inference.

| Supervised Approach

It is more consistent to use the supervised approach, in which we take these values as given and estimate the underlying sequence of latent states.

| Unsupervised Approach

We can also estimate the underlying transition probabilities and states, given that the number of states is known. This can be done with the Baum-Welch algorithm but can suffer from local minima in optimization.